

Henry Termondt

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Education

Campolindo High School (2022–Present)

Moraga, CA

GPA: Weighted: 4.53 / Unweighted: 4.0

SAT: 1600

Relevant Coursework: AP Computer Science A, AP Calculus AB, AP Calculus BC, AP Physics 1, AP Physics C, TA for AP Computer Science Principles

Foothill College (2023)

Los Altos Hills, CA

GPA: 4.0

Relevant Coursework: Introduction to Computer Forensics

Experiences/Activities

- **Independent Programming Projects** — To challenge myself, I researched & coded ideas not taught in school (partial differential equations, GPU acceleration, linear algebra). 50+ projects completed: games, fluid simulations, computer graphics algorithms and more. Fully self-taught using free resources. Almost all are open source. Earned top 0.05% popularity on Khan Academy CS community.
- **Helping Peers at School** — Provided daily support to several peers in all subjects; responded 24/7 to texts/calls. English, math, chemistry, science teachers asked for my assistance for students who had fallen behind. Gave guiding questions/explanations of concepts, never directly gave answers so they could learn to solve it themselves. Helped w/ studying, essay feedback, learning complex topics, catching up days absent.
- **Founder & President of Essential Life Skills Club** — Reached out to professionals in the community to give informational talks about important life skills (like taxes, credit/credit cards, and financial literacy). Coordinated with speakers to schedule events and do outreach. Recruited 57 members. I learned how to promote events and communicate w/ professionals. Bridged gap between school and real world.
- **Computational Physics Researcher** — Innovated/published novel way of implementing GPU-accelerated partial differential equation solvers, enhancing their accessibility. Developed open-source quantum mechanics simulator.
- **Violin** — Started in 4th grade, private lessons since 5th grade. Played jazz/folk solo w/ Diablo Symphony Orchestra and 2x Grammy-winning violinist Mads Tolling. Played in multiple recitals.

Played in high school orchestra every year. Convinced music teacher to add chamber music to curriculum so we all could learn how to play with others. Will play in school musical.

- **President of German Club** — Organized meetups, leadership meetings, and large events to share German culture/traditions. Managed budget/team to bring food/materials for events, attended by 60+ people, including faculty and the principal. Number of people attending events increased by ~30%. Collaborated w/ other clubs to add live music and other experiences. Delegated tasks to members. Got special permission for certain large events.
- **Chamber Music** — Independently organized a small group to play together. Scheduled and coordinated rehearsals with others. Performed in 6 concerts in retirement homes and in schools. Played 3 times for kids in music camps. Performed in a chamber orchestra.
- **Tutor for SAT Reading/Writing** — Tutored 5 students through Schoolhouse.world. Adapted teaching style to individual needs. Earned stellar reviews, leading to promotion to Senior tutor. Saw rewards of teaching.
- **Psilab Developer** — Built a website from scratch to help students visually understand wave functions in quantum mechanics and help teachers easily convey complex topics. I developed a technique for GPU acceleration of simulation to make it real-time. Used JavaScript, GLSL, HTML, and CSS. Promoted on open educational resource libraries: Merlot and OER Commons.
- **Game Design class at Stanford Pre-Collegiate Institute** — Analyzed game psychology, effects of player choice, industry operations and profitability. Built levels for and designed a game. ?
- **Assistant Concert Master of School Orchestra** — Auditioned Junior year for seating placement. Organized summer sectional meetup w/ Concert Master. Help lead sectionals, give orchestra constructive feedback on playing. Play solos when the Concert Master is unavailable. Explain bowings/fingerings and how to play more easily/musically to classmates. Shut down bullying of freshmen. Became a more confident performer.
- **Physics in and Through Cosmology Workshop** — Interviewed researchers at Berkeley Labs about career, experiences, and research. Attended lectures on cutting-edge particle physics/cosmology research. Met Nobel laureate Saul Perlmutter. Connected with talented high school students on physics topics. Worked with muon/cosmic ray detectors. Learned what it's like to be a researcher and pathway there.
- **Judge for Khan Academy Coding Competitions** — After being a contestant, I applied to become a judge for 5 competitions. Accepted based on the projects I made. Gave feedback on what was done well and what could be improved. Judged on code and code output (if the code is understandable/can be expanded on/its complexity; if the output is well designed/creative). Judged 30-40 per competition.
- **Codec Developer** — Curated, designed and programmed intermediate coding projects, fully explained to propel and motivate beginner programmers to more advanced levels.
- **Property Manager** — Hired by neighbors to take care of house/grounds for 3 summers and some holidays. Monitored for/addressed maintenance, security, and 3rd party vendor issues. Discovered/notified owners of pipe break before damage got too far. Trusted with access to their

house. Asked to return until they moved away. Vigilance/communication gave owners peace of mind.

Awards/Honors

- **International: 3rd Place in Nexus Journal Research Competition** — Developed a novel method for solving partial differential equations on small home computers in real time using GPU acceleration. Wrote an open-source sample implementation of a simulation of the 2D time-independent Schrödinger equation (used in quantum mechanics). Wrote a research paper on the method and results. Won 3rd place and was published.
- **National: AP Scholar with Distinction Award** — I received six 5's on the AP exams I took before senior year. Subjects were AP Computer Science A, AP Calculus AB, AP Physics 1, AP English Language and Composition, AP Biology, and AP United States History.
- **State: Certificate of Merit, Levels 3, 4, & 5 with State Honors from the Music Teachers' Association of CA** — Took music theory tests (covered ear training/notation/intervals), played 2 concertos, an etude, and 3 scales. Scored 68/69, 75/75, 75/75 on music theory tests and awarded perfect scores on all performances. Recorded performances and submitted. For concertos, practiced/played with piano accompanists. Played at high level leading to state honors.
- **State: California State Seal of Biliteracy** — Took German all four years of high school and have received an A every semester.
- **School: Campolindo Service Award** — Did community service for 40–50 hours my sophomore and junior years.

Skills

Programming

- Proficient in JavaScript, HTML, CSS, Python, Java, and GLSL
- Intermediate in Rust and WebAssembly
- Proficient in making websites, complex physics simulations, GPU programming and designing games

Communication

- Presenting
- Poetry-writing — Three haiku accepted for publication by the Scientific American, a magazine with 10 million monthly readers

German Language

- Proficient

Music

- Assistant Concertmaster of Campolindo Orchestra, 2nd chair of 1st violins (2025–Present)
- Student soloist of Diablo Symphony Orchestra (2024)